

## 04.03 Applying Roots and Powers

Common problems and solution methods

### Rule guide: recognize the task first

| Recognize                           | Apply these rules                                  |
|-------------------------------------|----------------------------------------------------|
| A power applies to a product.       | Distribute the power to every factor.              |
| A power applies to a quotient.      | Apply the power to both numerator and denominator. |
| A power is raised to another power. | Multiply the exponents.                            |

### Example 1: Apply a power to a product

|         |            |
|---------|------------|
| Problem | $(2x^3)^2$ |
|---------|------------|

#### How to solve it

Distribute the power:

$$(2x^3)^2 = 2^2(x^3)^2$$

Simplify:

$$= 4x^6$$

### Example 2: Apply a power to a quotient

|         |                                |
|---------|--------------------------------|
| Problem | $\left(\frac{3x}{2y}\right)^2$ |
|---------|--------------------------------|

#### How to solve it

Square numerator and denominator:

$$\left(\frac{3x}{2y}\right)^2 = \frac{(3x)^2}{(2y)^2}$$

Simplify:

$$= \frac{9x^2}{4y^2}$$

### Example 3: Raise a power to a power

|         |           |
|---------|-----------|
| Problem | $(x^4)^3$ |
|---------|-----------|

#### How to solve it

Multiply exponents:

$$(x^4)^3 = x^{4 \cdot 3}$$

Simplify:

$$= x^{12}$$